

05 Inverse of a Matrix

5.2 Properties of the Matrix Inverse

Why matrices fail to invert, the inverse of a product, and the inverse under diagonal, symmetric, transpose, orthogonal, kA , and determinant

Linear Algebra Made Easy and Visual | Open-source code & drafts: <https://github.com/Visualize-ML>

Core Skills You Will Build

1

Build geometric intuition: understand these properties through projection, dimension collapse, and composite transforms

3

The geometric meaning of a non-invertible square matrix: lost information, such as dimension collapse or projection

5

The inverse of a matrix product: the inverse of a product of invertible matrices reverses the order of the individual inverses

7

The inverse of a symmetric matrix is itself symmetric

2

Test invertibility: a nonzero determinant is the necessary and sufficient condition for a square matrix to be invertible

4

A non-square matrix is always non-invertible: mismatched input/output dimensions make the original vector unrecoverable

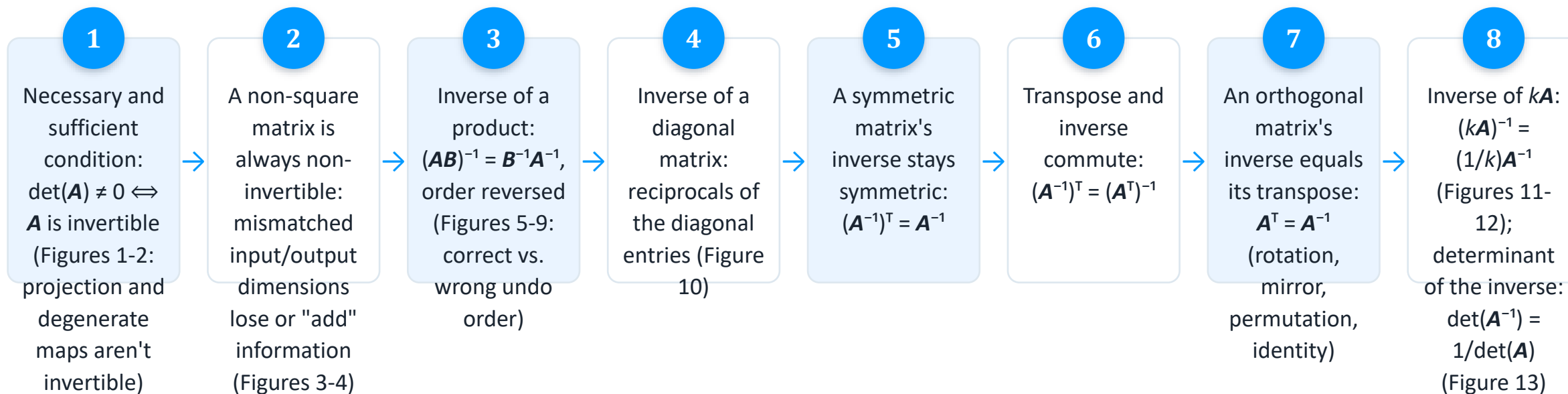
6

The inverse of an invertible diagonal matrix: reciprocals of the diagonal entries

8

The inverse of an orthogonal matrix equals its transpose

How This Section Unfolds



A Nonzero Determinant Is Necessary and Sufficient for Invertibility

$$\mathbf{P} = \begin{bmatrix} 1/4 & -\sqrt{3}/4 \\ -\sqrt{3}/4 & 3/4 \end{bmatrix} \Rightarrow \det(\mathbf{P}) = 0$$

- "Necessary and sufficient" means: the condition guarantees the conclusion (sufficiency), and the conclusion guarantees the condition (necessity)
- Given the 2×2 symmetric matrix $\mathbf{P}$ above, its determinant works out to 0, so $\mathbf{P}$ is not invertible
- To see why $\mathbf{P}$ is not invertible, we can analyze its effect as a projection

Figure 1 · A Projection in the Plane Isn't Invertible

- The 2×2 symmetric matrix $\mathbf{P}$ projects any planar shape onto a line through the origin, losing information that can't be recovered
- $\mathbf{P}$'s two column vectors satisfy $\mathbf{p}_2 = -\sqrt{3} \cdot \mathbf{p}_1$, so they're collinear — the zero-area parallelogram they span matches $\det(\mathbf{P}) = 0$

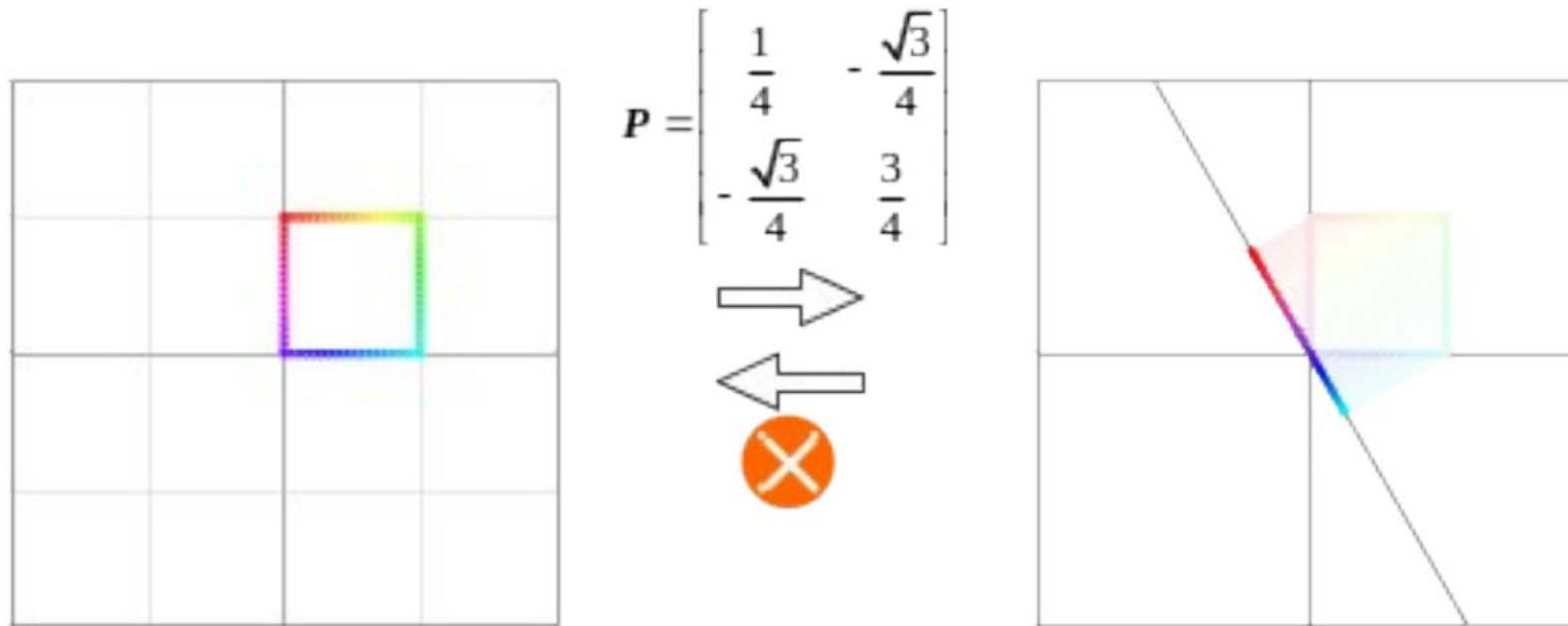
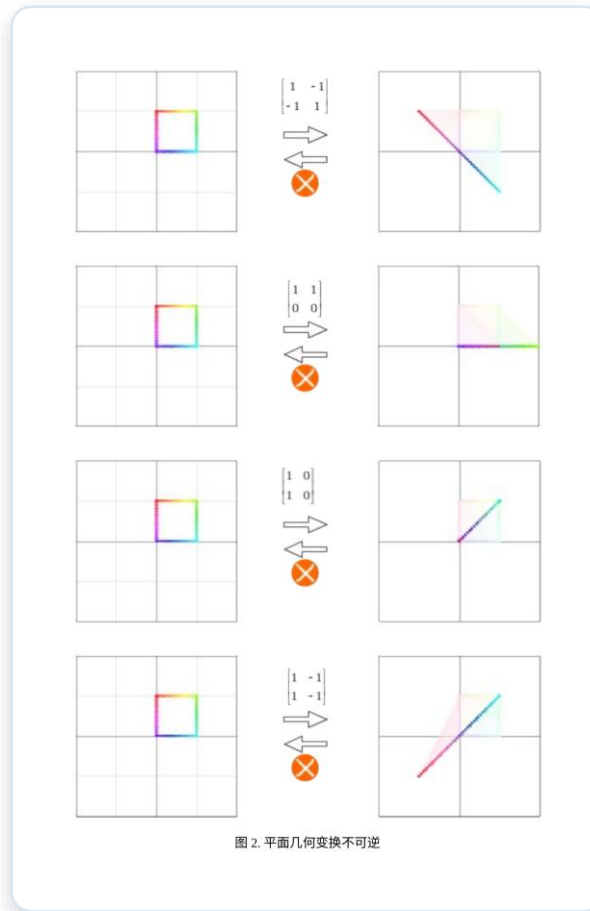


图 1. 平面上的投影不可逆, 2×2 矩阵, 第一例

Figure 2 · Planar Geometric Transforms That Aren't Invertible

- Each of these four 2x2 matrices squashes the unit square onto a single line through the origin — all four transforms are non-invertible
- Try analyzing each matrix's determinant yourself, and check whether its column vectors are linearly dependent



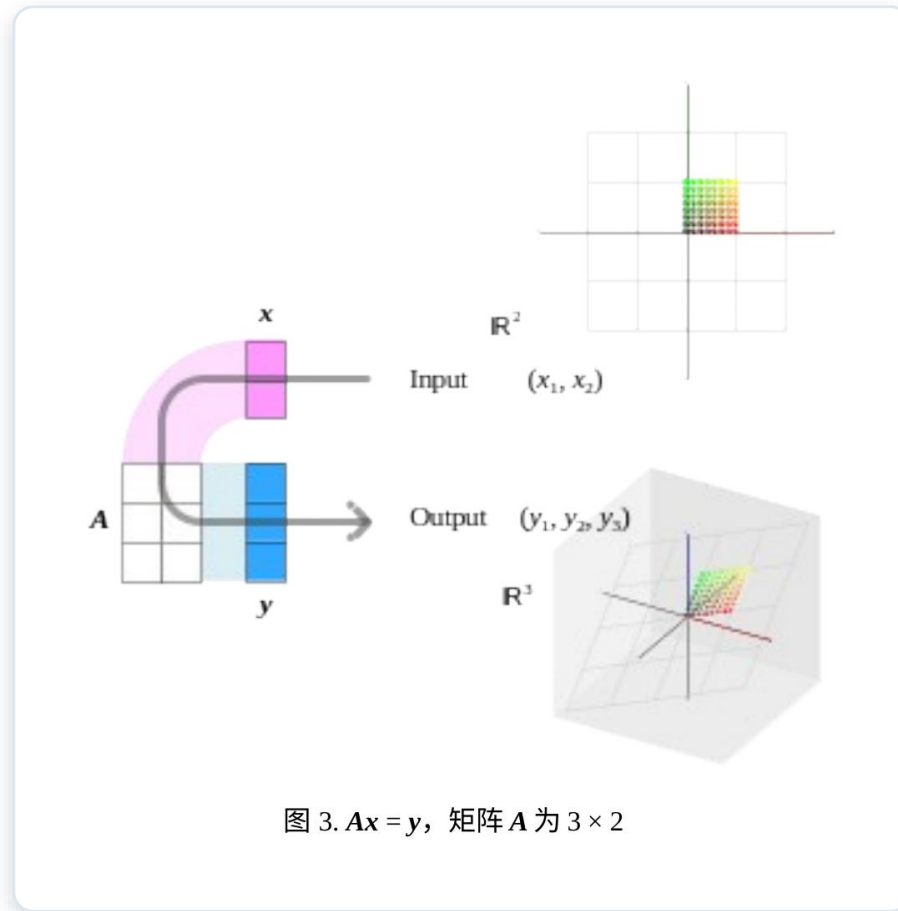
A Non-square Matrix Is Always Non-invertible

$$\mathbf{Ax} = [a_{11} \ a_{12}; a_{21} \ a_{22}; a_{31} \ a_{32}][x_1; x_2] = [y_1; y_2; y_3]$$

- If a matrix isn't square — rows $\neq$ columns — it's not invertible: mismatched input and output dimensions mean the original input can't be uniquely recovered
- As shown above, the 3×2 matrix $\mathbf{A}$ maps 2D space into some 2D plane sitting inside 3D space, but never covers all of 3D space

Figure 3 · $Ax = y$, Matrix A Is 3×2

- Information seems to "increase" ($2D \rightarrow 3D$), yet the original 2D coordinates can't be recovered — the extra dimension is free to vary
- Picture points drawn on paper (2D) placed into 3D space — they still lie on one plane, so 2D coordinates can't be recovered



Losing a Dimension Isn't Invertible Either

$$\mathbf{A}\mathbf{x} = \begin{bmatrix} b_{11} & b_{12} & b_{13} \\ b_{21} & b_{22} & b_{23} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} y_1 \\ y_2 \end{bmatrix}$$

- The 2×3 matrix $\mathbf{A}$ maps 3D space into 2D space: the mapping loses information, so different 3D vectors can project onto the same 2D vector
- A non-square matrix is always non-invertible — mismatched input and output dimensions mean information is lost or gained, so the inverse transform can never uniquely recover the original vector
- Non-invertible square matrices and non-square matrices still have a kind of "inverse" — the pseudoinverse, which minimizes error and is used to solve least-squares problems; this is covered later in the book

Figure 4 · $Ax = y$, Matrix A Is 2×3

- Picture standing in a room (3D), looking at an object's shadow on the wall (2D) — the shadow alone can't reveal the full 3D shape
- That's why a 2×3 matrix isn't invertible: going from 3D to 2D loses information, so the inverse map isn't unique

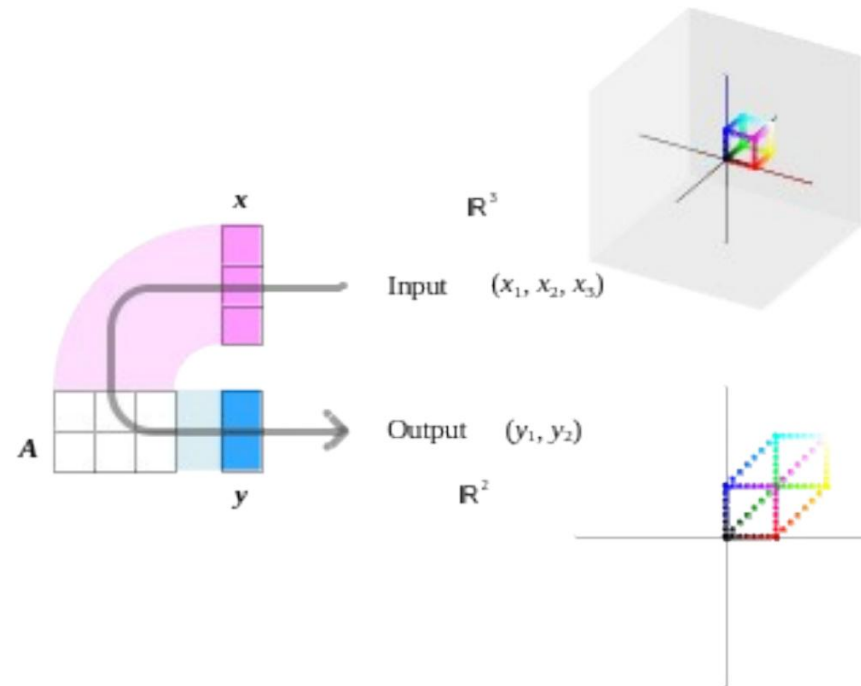


图 4. $Ax = y$, 矩阵 A 为 2×3

$(AB)^{-1} = B^{-1}A^{-1}$, the Order Reverses

$$(AB)^{-1} = B^{-1}A^{-1}$$

- If same-shaped square matrices A and B are both invertible, then AB is invertible too, and its inverse is $B^{-1}A^{-1}$ — the order is reversed from the original product
- Geometrically, $A @ B @ x$ means: for vector x , first apply B 's geometric operation, then apply A 's
- Using our earlier examples: $R = [-1/2 \ -\sqrt{3}/2; \sqrt{3}/2 \ -1/2]$ is a rotation matrix, and $S = [3/2 \ 0; 0 \ 2/3]$ is a scaling matrix

Figure 5 · A Composite Transform

- $((R @ S) @ x)$ represents a "one-shot" composite geometric transform — the two operations S and R are merged into a single matrix multiplication
- This is exactly what makes matrix multiplication so powerful: several geometric transforms can be pre-composed into one matrix and then applied all at once

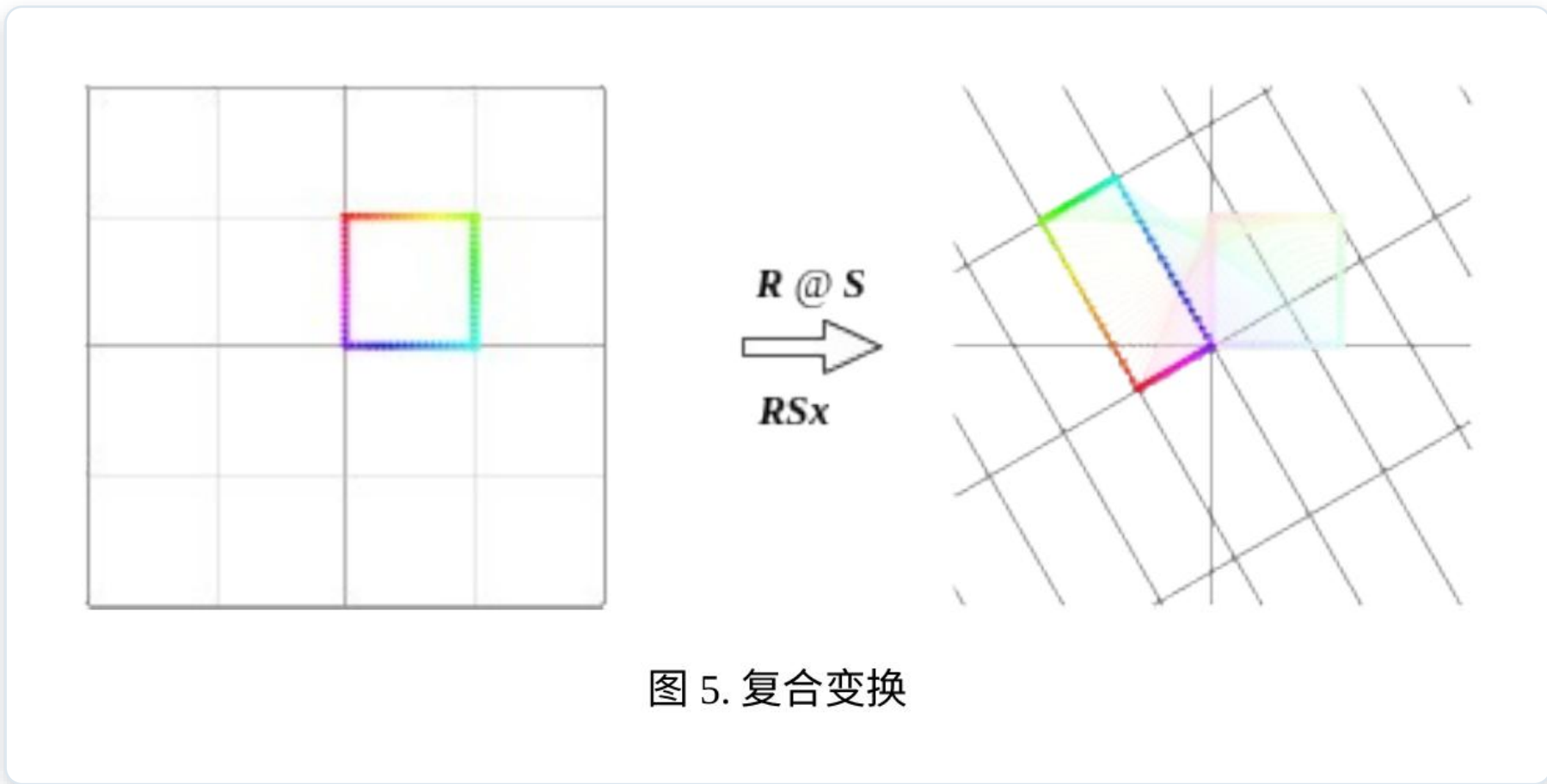


图 5. 复合变换

Figure 6 · The Inverse of a Composite Transform

- The inverse of $(R @ S)$ is $(R @ S)^{-1}$, so $(R @ S)^{-1} @ (R @ S) @ x = I @ x = x$
- The inverse of a composite transform can likewise restore the figure back to the original unit square in one shot

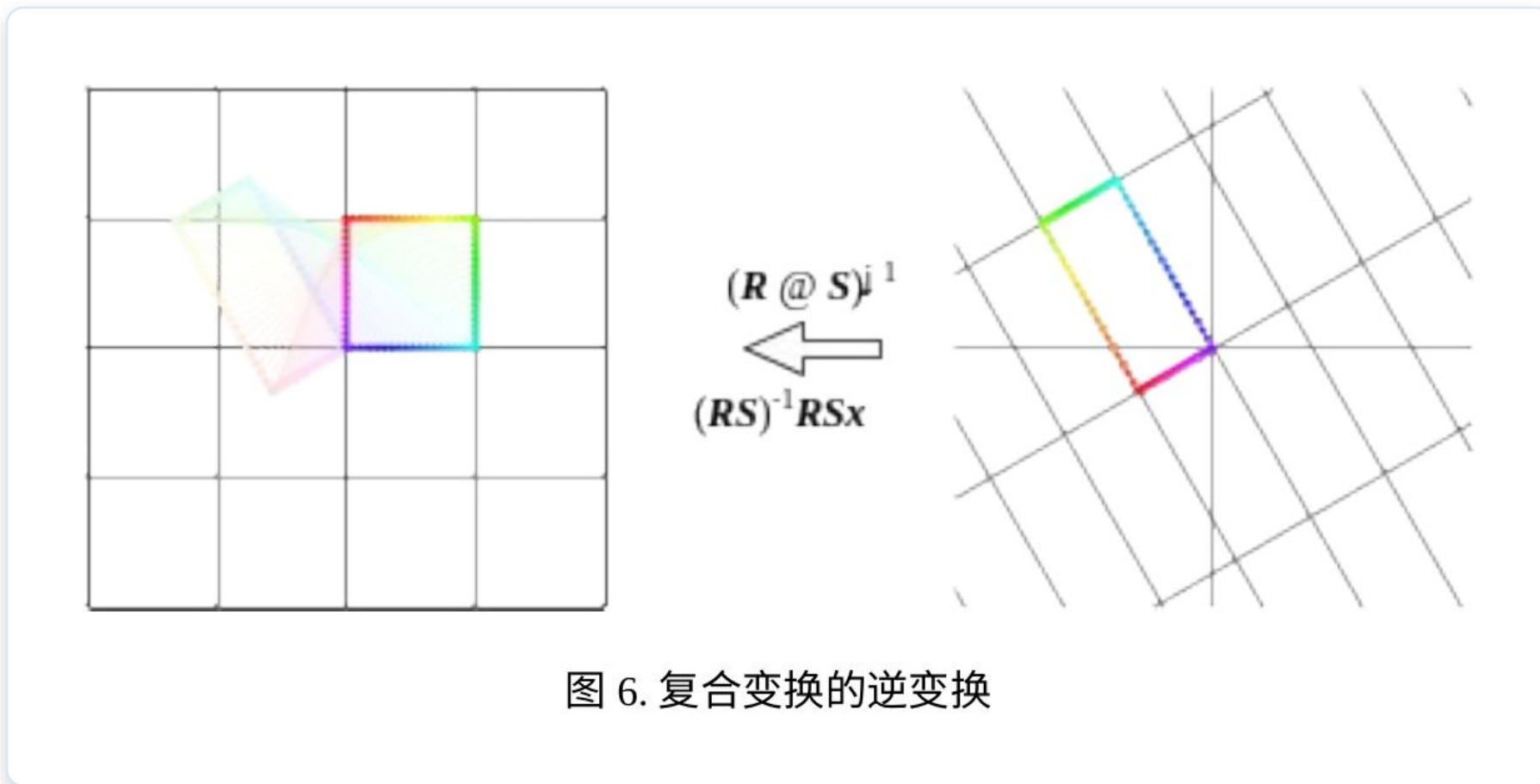


图 6. 复合变换的逆变换

Expanding the Composite: Scale First, Then Rotate

$$\mathbf{R} @ \mathbf{S} @ \mathbf{x} = \text{Rotate}(\mathbf{R}) @ \text{Scale}(\mathbf{S}) @ \mathbf{x}$$

- Expanding the composite operation $((\mathbf{R} @ \mathbf{S}), \mathbf{S}$ scales first, then $\mathbf{R}$ rotates — "scale first, then rotate"
- In a matrix product $\mathbf{A} @ \mathbf{B}$, the matrix closest to the vector $\mathbf{x}$ — here $\mathbf{B}$ — acts first; this matters a great deal when expanding a composite transform

Figure 7 · Scale First, Then Rotate

- The unit square is first scaled by S to get Sx , then rotated by R to get RSx — these two steps are equivalent to the single composite transform $(R @ S) @ x$

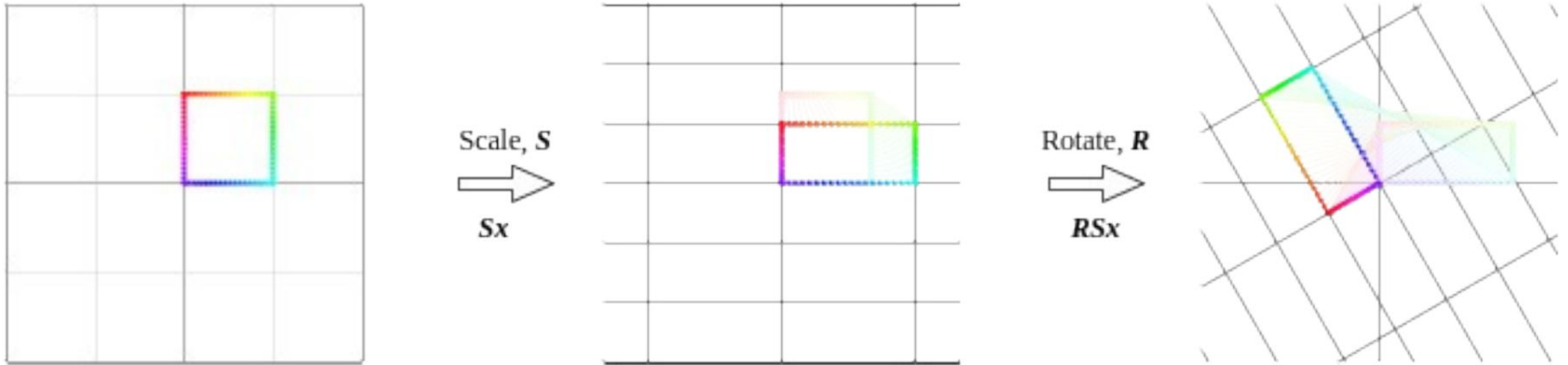


图 7. 先缩放、再旋转

Undoing It: Remove R First, Then Remove S

$$S^{-1} @ R^{-1} @ R @ S @ x = I @ x = x$$

- Working backward, we must first undo R — the operation applied last — and only then undo S ; this is exactly where $(RS)^{-1} = S^{-1}R^{-1}$ comes from geometrically
- The order can't be swapped: each geometric step must be undone in "last in, first out" order

Figure 8 · Inverse Transform: Rotate Back First, Then Scale Back

- Undoing in order: first apply R^{-1} to reverse the rotation from R , then apply S^{-1} to reverse the scaling from S
- Once both inverse steps are applied in the correct order, the figure is fully restored to the original unit square

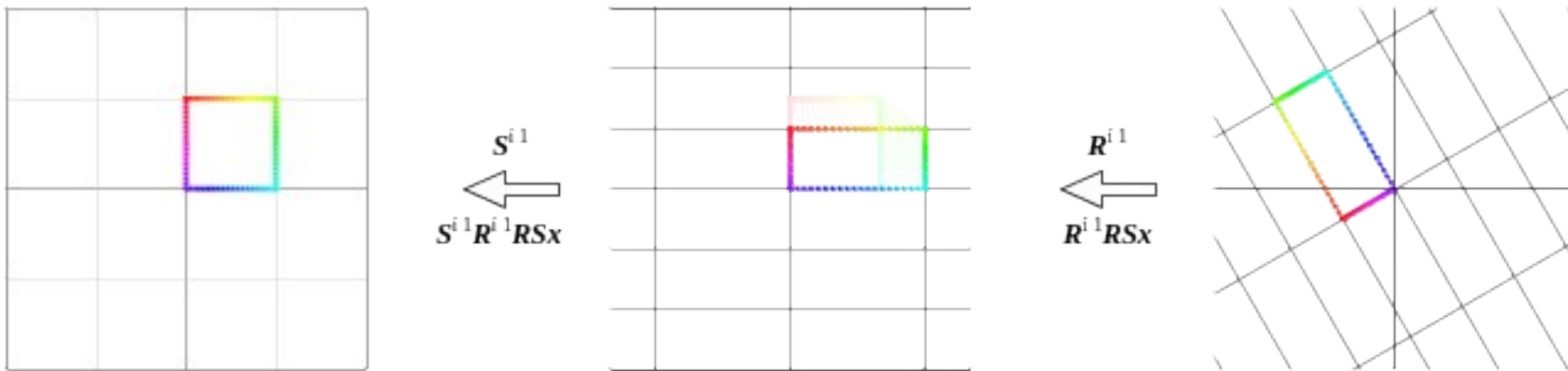


图 8. 逆变换：先反向旋转、再反向缩放

What If the Order Is Reversed?

$$\mathbf{R} @ \mathbf{S} @ \mathbf{R}^{-1} @ \mathbf{S}^{-1} @ \mathbf{x} \neq \mathbf{x}$$

- What happens if we reverse the scaling first ($\mathbf{S}^{-1}$) and then reverse the rotation ($\mathbf{R}^{-1}$) — that is, we don't undo the steps in "last in, first out" order?
- Figure 9, next, shows this wrong-order inverse transform, making it clear why reversing a product's inverse must strictly flip the order

Figure 9 · The Wrong Inverse Transform: Scale Back First, Then Rotate Back

- As expected, undoing the steps in the wrong order fails to restore the original figure (the unit square) — which is exactly why $(AB)^{-1}$ must equal $B^{-1}A^{-1}$, not $A^{-1}B^{-1}$
- Only by strictly "undoing the later operation first, then the earlier one" can the inverse transform succeed

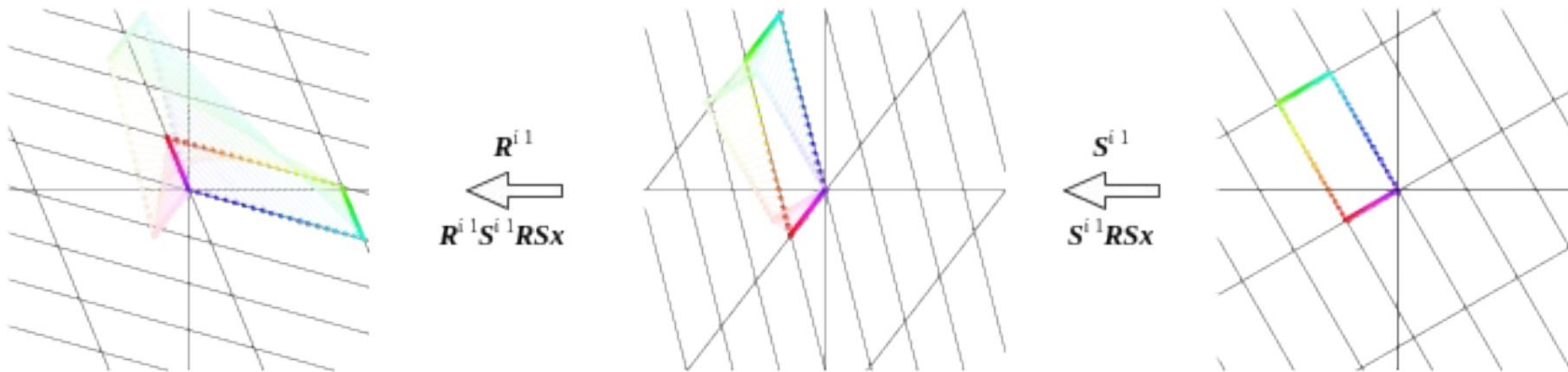


图 9. 错误的逆变换：先反向缩放、再反向旋转

A Diagonal Matrix's Inverse Takes Reciprocals of the Diagonal

$$\mathbf{S} = \begin{bmatrix} 2/3 & 0 \\ 0 & 3/2 \end{bmatrix} \Rightarrow \mathbf{S}^{-1} = \begin{bmatrix} 3/2 & 0 \\ 0 & 2/3 \end{bmatrix}$$

- A diagonal matrix's role: each diagonal entry is an independent scaling factor, applied along the x_1 -axis and x_2 -axis respectively
- $\mathbf{S}$'s inverse is literally the reciprocal of each diagonal entry (note: this $\mathbf{S}$ uses different numbers from the $\mathbf{S}$ in the earlier composite-transform example — it's a separate, standalone example)
- As long as every diagonal entry is nonzero, a diagonal matrix is guaranteed to be invertible

Figure 10 · Scaling in the Plane, and Its Inverse

- The square is first scaled independently along each coordinate axis by S , then transformed by S^{-1} , fully restoring the original square

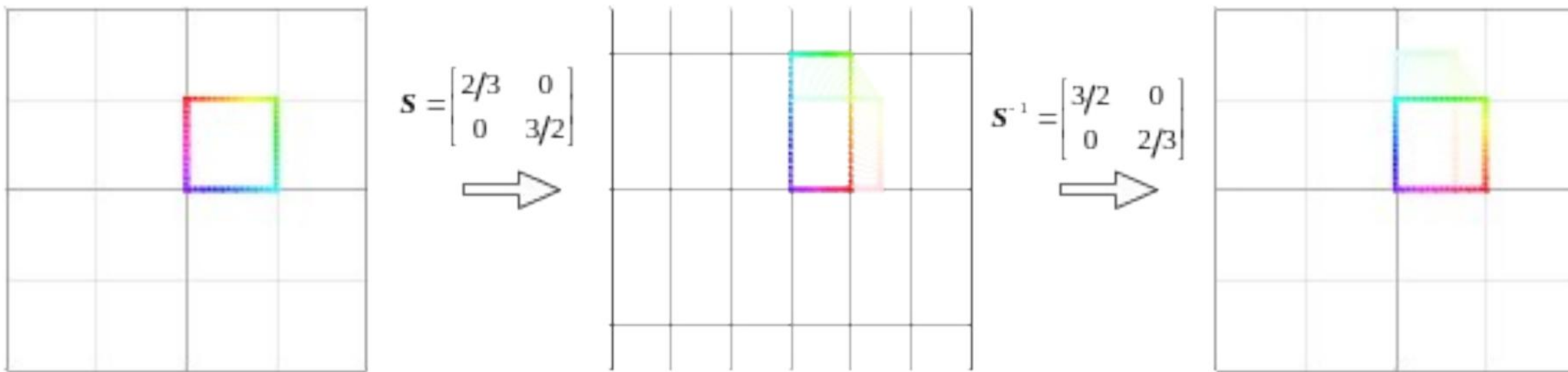


图 10. 平面上的缩放、缩放的逆操作

A Symmetric Matrix's Inverse Stays Symmetric

$$\text{If } \mathbf{A} = \mathbf{A}^T, \text{ then } (\mathbf{A}^{-1})^T = \mathbf{A}^{-1}$$

- A short proof: $(\mathbf{A}^{-1})^T \mathbf{A} \mathbf{A}^{-1} = (\mathbf{A}^{-1})^T \mathbf{A}^T \mathbf{A}^{-1} = (\mathbf{A} \mathbf{A}^{-1})^T \mathbf{A}^{-1} = \mathbf{A}^{-1}$
- In other words, if $\mathbf{A}$ is symmetric, its inverse $\mathbf{A}^{-1}$ is guaranteed to be symmetric too

The Inverse of a Transpose Equals the Transpose of an Inverse

$$(\mathbf{A}^{-1})^T = (\mathbf{A}^T)^{-1}$$

- A short proof: $(\mathbf{A}^{-1})^T = (\mathbf{A}^{-1})^T \mathbf{A}^T (\mathbf{A}^T)^{-1} = ((\mathbf{A}^{-1})^T \mathbf{A}^T) (\mathbf{A}^T)^{-1} = (\mathbf{A} \mathbf{A}^{-1})^T (\mathbf{A}^T)^{-1} = (\mathbf{A}^T)^{-1}$
- The geometric intuition behind this identity relies on the Singular Value Decomposition (SVD) — one of the important topics covered later in this book

An Orthogonal Matrix's Inverse Equals Its Transpose

$$\mathbf{A}^T = \mathbf{A}^{-1} \implies \mathbf{A}^T \mathbf{A} = \mathbf{A} \mathbf{A}^T = \mathbf{I}$$

- A rotation matrix is one type of orthogonal matrix — for example $\mathbf{V} = [\cos\vartheta \ -\sin\vartheta; \sin\vartheta \ \cos\vartheta]$ rotates the plane counterclockwise about the origin by ϑ
- $\mathbf{V}$'s inverse is $\mathbf{V}^{-1} = \mathbf{V}^T = [\cos\vartheta \ \sin\vartheta; -\sin\vartheta \ \cos\vartheta]$; both $\mathbf{V} @ \mathbf{V}^T$ and $\mathbf{V}^T @ \mathbf{V}$ equal the identity matrix
- Worth noting: the identity matrix, mirror matrices, and permutation matrices are all orthogonal matrices too

Inverting $k\mathbf{A}$ Needs an Extra Reverse Scale by $1/k$

$$(k\mathbf{A})^{-1} = (1/k)\mathbf{A}^{-1}, k \neq 0$$

- If square matrix $\mathbf{A}$ is invertible, then $k\mathbf{A}$'s inverse is $(1/k)\mathbf{A}^{-1}$ — besides applying $\mathbf{A}^{-1}$, the inverse also needs a reverse scale by $1/k$ to cancel out k 's effect
- Writing an $n \times n$ matrix $\mathbf{A}$ as a set of row vectors, $k\mathbf{A}$ is the same as left-multiplying $\mathbf{A}$ by the diagonal matrix $\text{diag}(k, \dots, k)$, which scales every row of $\mathbf{A}$

Figure 11 · Scaling Every Row of a Flat Matrix A

- Left-multiplying a flat matrix A (few rows, many columns) by a diagonal matrix S scales every row of A — if the diagonal entries differ, each row gets its own scaling factor

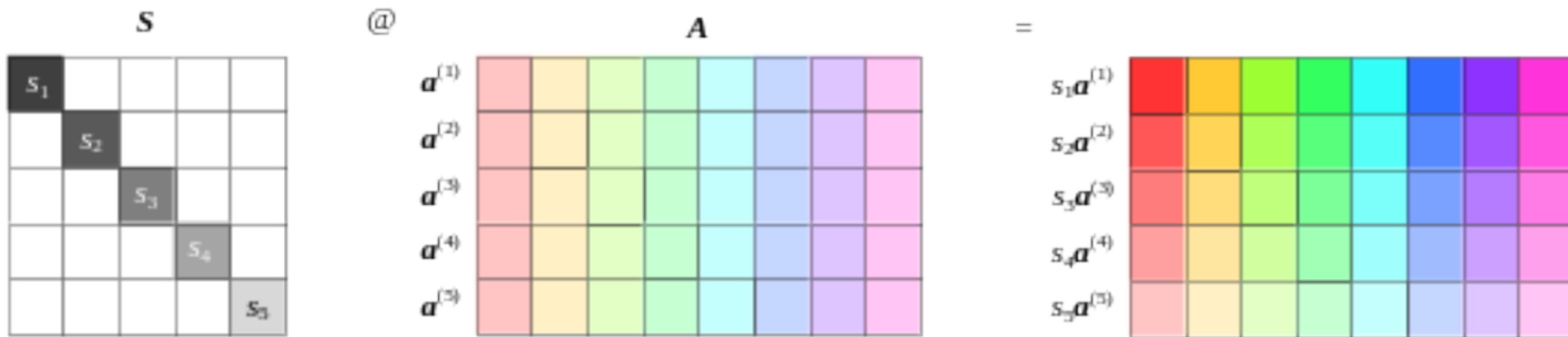


图 11. 对扁平矩阵 A 的每行缩放

Figure 12 · Scaling Every Column of a Tall, Thin Matrix $\mathbf{A}$

- Conversely, writing an $n \times n$ matrix $\mathbf{A}$ as a set of column vectors, right-multiplying $\mathbf{A}$ by a diagonal matrix $\mathbf{S}$ scales every column of $\mathbf{A}$
- Note that $\mathbf{A}$'s shape differs here: flat (few rows) in Figure 11, tall and thin (few columns) in Figure 12

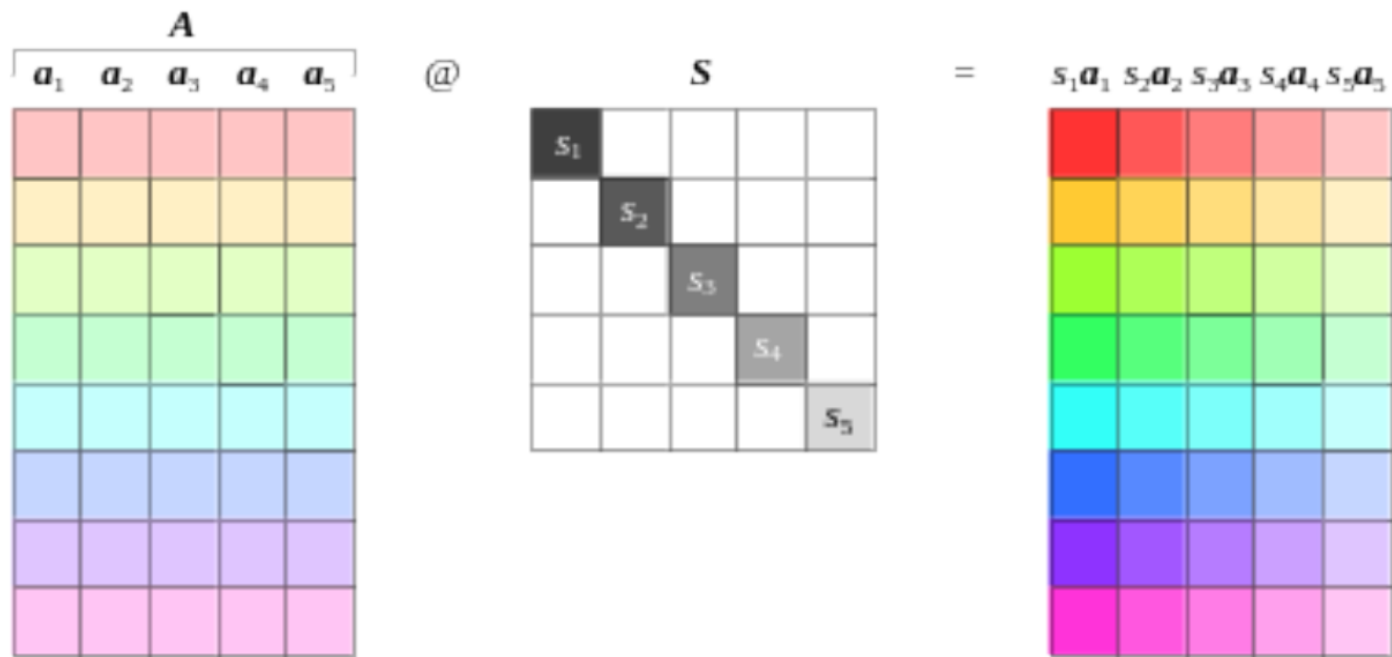


图 12. 对细高矩阵 $\mathbf{A}$ 的每列缩放

$$\det(\mathbf{A}^{-1}) = 1 / \det(\mathbf{A})$$

$$\det(\mathbf{A}^{-1}) = 1 / \det(\mathbf{A})$$

- A matrix $\mathbf{A}$'s determinant $\det(\mathbf{A})$ is the scaling factor it applies to area or volume; if $\mathbf{A}$ is invertible, its inverse's determinant is exactly the reciprocal of $\det(\mathbf{A})$
- As shown in Figure 13, if $\mathbf{A}$ doubles the area of the unit square — $\det(\mathbf{A}) = 2$ — then its inverse $\mathbf{A}^{-1}$ shrinks the area back to $1/2$, so $\det(\mathbf{A}^{-1}) = 1/2$

Figure 13 · Scaling of Area

- The unit square's area doubles after the transform $\mathbf{A}$; applying $\mathbf{A}^{-1}$ then shrinks the area back to half, exactly restoring the unit square

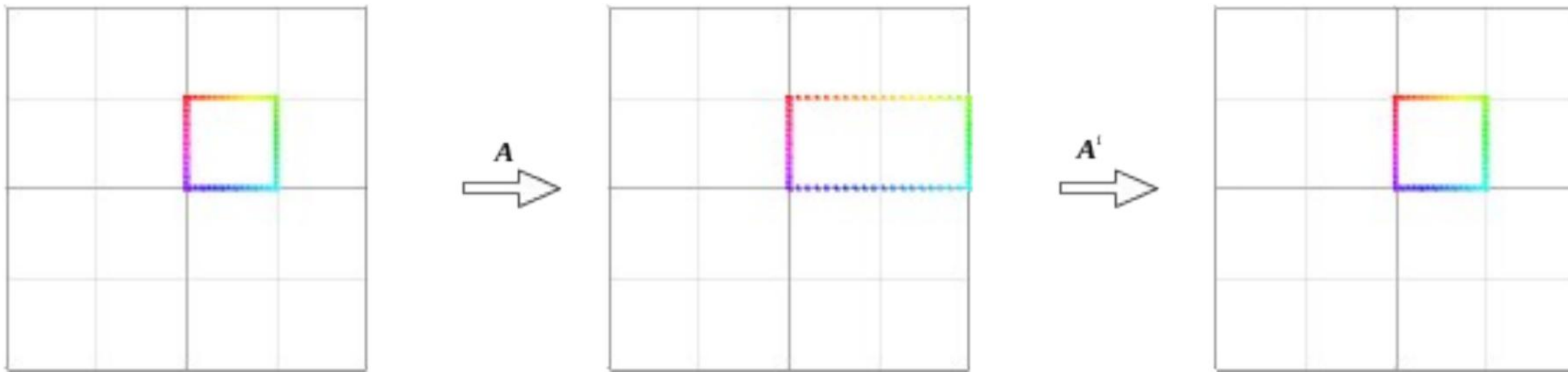


图 13. 面积的缩放

The Inverse's Properties All Interlock

$$\det(A^{-1}) = 1/\det(A), (AB)^{-1} = B^{-1}A^{-1}$$

- Invertibility hinges on $\det(A) \neq 0$; a non-square matrix is never invertible; inverting a product reverses the order — all of this traces back to whether a geometric transform can be uniquely undone
- Diagonal, symmetric, and orthogonal matrices each have an elegant inverse structure: reciprocal diagonal entries, preserved symmetry, and transpose-equals-inverse
- Next, we'll learn how to solve for the inverse of a general square matrix by hand

Key Terms in This Section

Necessary and Sufficient Condition $\det(A) \neq 0$ and A being invertible imply each other

Pseudoinverse A generalized inverse for non-square or singular matrices, used for least-squares problems, covered later

Inverse of a Diagonal Matrix Reciprocal of each nonzero diagonal entry; still diagonal

Inverse of a Transpose $(A^{-1})^T = (A^T)^{-1}$ — transpose and inverse commute

Inverse of kA $(kA)^{-1} = (1/k)A^{-1}$ ($k \neq 0$) — equivalent to scaling every row or column of A by a diagonal matrix $\text{diag}(k, \dots, k)$

Non-invertible / Singular Matrix A square matrix with zero determinant; its geometric transform loses information

Inverse of a Product $(AB)^{-1} = B^{-1}A^{-1}$ — order reversed, mirroring "last in, first out" when undoing geometric steps

Inverse of a Symmetric Matrix If $A = A^T$, then $(A^{-1})^T = A^{-1}$ — the inverse stays symmetric

Inverse of an Orthogonal Matrix $A^T = A^{-1}$ — rotation, mirror, permutation, and identity matrices are all orthogonal

Determinant of the Inverse $\det(A^{-1}) = 1/\det(A)$ — the area/volume scaling factors of a transform and its inverse are reciprocals

Companion Code and Further Reading

Further Reading · Pseudoinverse and Least Squares

The pseudoinverse for non-square or singular matrices, and its use in least-squares problems — covered later in this book

Further Reading · Singular Value Decomposition (SVD)

The key tool for understanding the geometry behind $(A^{-1})^T = (A^T)^{-1}$ — introduced systematically later in this book

Further Reading · numpy Linear Algebra Tools

`numpy.linalg.matrix_rank()`, `numpy.linalg.pinv()`, `numpy.linalg.cond()` — explore these on your own

Open-source Repository

All code and figures for this book: <https://github.com/Visualize-ML>

Q1 – Q8

Q1 Which of the following matrices are orthogonal? ① $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ ② $\begin{bmatrix} 1 & 0 \\ 0 & 2 \end{bmatrix}$ ③ $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ ④ $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ ⑤ $\begin{bmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{bmatrix}$ ⑥ $\begin{bmatrix} 3 & -4 \\ 4 & 3 \end{bmatrix}$.

Q3 Identify which of these square matrices are non-invertible, and explain your reasoning: ① $\begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$ ② $\begin{bmatrix} 1 & -1 \\ -1 & 1 \end{bmatrix}$ ③ $\begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$ ④ $\begin{bmatrix} 1 & 1 \\ 1 & 1 \end{bmatrix}$.

Q5 Use `numpy.linalg.matrix_rank()` to compute the rank of each matrix in Q3, and explain the relationship between rank and invertibility.

Q7 Use `numpy.linalg.pinv()` to compute the pseudoinverse of the 3×2 matrix A from Figure 3, and check whether $A @ \text{pinv}(A)$ and $\text{pinv}(A) @ A$ equal the identity matrix.

Q2 Compute the inverse of each matrix below and explain its geometric meaning: ① $\begin{bmatrix} 4 & 0 \\ 0 & 2 \end{bmatrix}$ ② $\begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix}$ ③ $\begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix}$ ④ $\begin{bmatrix} 3/5 & -4/5 \\ 4/5 & 3/5 \end{bmatrix}$.

Q4 Write a function `is_invertible(A)` that checks whether a square matrix is invertible by testing its determinant, and use it to verify the 6 matrices in Q1.

Q6 Write a function `try_invert(A)` that calls `numpy.linalg.inv()` on a non-square matrix (e.g. 3×2), and observe and explain the error.

Q8 Construct two invertible 2×2 matrices A and B, verify with code that $(AB)^{-1} = B^{-1}A^{-1}$, and show that $(AB)^{-1}$ generally does not equal $A^{-1}B^{-1}$.

Q9 – Q16

Q9 Write a function `compose_order_matters(A, B, x)` that computes $A @ B @ x$ and $B @ A @ x$ separately, and visualize how matrix multiplication fails to commute.

Q11 Verify with code that Figure 9's "reverse-scale then reverse-rotate" fails to restore the original square, and compute the restoration error (e.g. the Frobenius norm).

Q13 Write functions `is_symmetric(A)` and `is_orthogonal(A)` that check whether a matrix is symmetric or orthogonal, and use them on every matrix that appears in this section.

Q15 Use SymPy symbolic computation to verify: $(A^{-1})^T = (A^T)^{-1}$ (the inverse of a transpose equals the transpose of an inverse).

Q10 Using this section's rotation matrix R and scaling matrix $S = \begin{bmatrix} 3/2 & 0 \\ 0 & 2/3 \end{bmatrix}$, reproduce Figures 7 and 8, verifying whether "scale then rotate" and "reverse-rotate then reverse-scale" restore the unit square.

Q12 Construct a 3×3 diagonal matrix $D = \text{diag}(2, 4, 5)$, compute its inverse by hand, and verify with `numpy.linalg.inv()`.

Q14 Use SymPy symbolic computation to verify: if $A = A^T$, then $(A^{-1})^T = A^{-1}$ (a symmetric matrix's inverse stays symmetric).

Q16 Write a function `rotation_matrix(theta)` that returns a 2×2 rotation matrix V , and verify that both $V @ V.T$ and $V.T @ V$ equal the identity matrix.

Q17 – Q23

Q17 Besides rotation matrices, use code to verify whether the identity matrix, the mirror matrix $[-1,0;0,1]$, and the permutation matrix $[0,1;1,0]$ all satisfy the definition of an orthogonal matrix.

Q19 Write a function `scale_cols(A, D)` that right-multiplies matrix A by diagonal matrix D , and verify the result equals scaling each column of A by D 's diagonal entries (matching Figure 12).

Q21 Write a function `det_inverse_check(A)` that verifies whether $\det(A^{-1})$ equals $1/\det(A)$ (tolerance $1e-9$).

Q23 Write a function `matrix_summary(A)` that reports whether A is invertible, symmetric, and orthogonal, along with its determinant and inverse (if it exists), summarizing everything covered in this section.

Q18 Write a function `scale_rows(D, A)` that left-multiplies matrix A by diagonal matrix D , and verify the result equals scaling each row of A by D 's diagonal entries (matching Figure 11).

Q20 Construct an invertible matrix A and a scalar $k=3$, and verify with code whether $(kA)^{-1}$ equals $(1/k)A^{-1}$.

Q22 Construct a 2×2 matrix A with determinant 2, use matplotlib to visualize how the unit square's area changes under A and A^{-1} , and verify the area scaling factors are 2 and $1/2$ respectively.

10 Prompts to Explore Further (1/2)

- 1 Explain why a nonzero determinant is both a sufficient and a necessary condition for a matrix to be invertible.

- 2 Explain why a non-square matrix is always non-invertible, and how that connects to input and output dimensions being unequal.

- 3 Explain why $(AB)^{-1} = B^{-1}A^{-1}$ rather than $A^{-1}B^{-1}$, using the order of geometric transforms.

- 4 Using the examples in Figures 7, 8, and 9, explain why undoing a composite geometric transform must follow strict "last in, first out" order.

- 5 Explain why the inverse of a diagonal matrix is obtained simply by taking the reciprocal of each diagonal entry.

10 Prompts to Explore Further (2/2)

6

Explain why the inverse of a symmetric matrix is always symmetric.

7

Explain the identity $(A^{-1})^T = (A^T)^{-1}$ — why can transpose and inverse be swapped?

8

Explain why an orthogonal matrix's inverse equals its transpose, and what these matrices have in common geometrically.

9

Explain why $(kA)^{-1} = (1/k)A^{-1}$, and how that connects to scaling every row or column of a matrix.

10

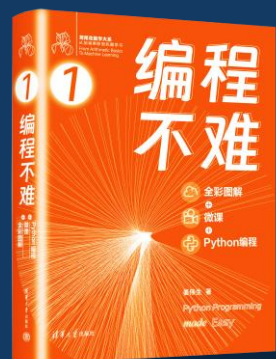
Explain why $\det(A^{-1}) = 1/\det(A)$, and how that connects to the scaling factor for area or volume.

数学

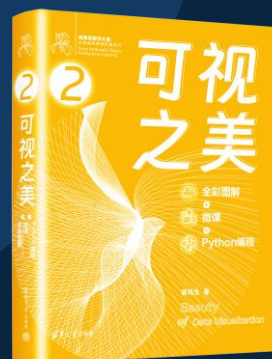
数学基础

线性代数

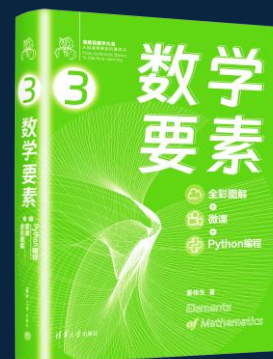
概率统计



Python编程



数据可视化



工具

数据分析
图和网络

回归、降维
分类、聚类

实践

